$$\frac{\mathrm{d}}{\mathrm{d}t}\hat{\rho}_{\mathcal{S}}(t) = -\mathrm{i}\big[\hat{H}'_{\mathcal{S}}, \hat{\rho}_{\mathcal{S}}(t)\big] - \mathrm{i}\gamma_0\big[\hat{X}, \big\{\hat{P}, \hat{\rho}_{\mathcal{S}}(t)\big\}\big] - 2M\gamma_0 k_{\mathrm{B}}T\big[\hat{X}, \big[\hat{X}, \hat{\rho}_{\mathcal{S}}(t)\big]\big]. \tag{5.203}$$